

What we're doing

Can we train the exp117 Qwen3 1.47B recipe directly from the reusable ESMFold2 Atlas pyconfind-contact rows, constructing the canonical `contacts-v1` documents and token sequences at read time rather than materializing a text corpus?

Why

A shard-local direct Levanter dataset can reconstruct `AnalyzedStructure`, call the existing deterministic `contacts-v1` builder, encode with the pinned contacts tokenizer vocabulary, and greedily pack the same document stream quickly enough to feed a TPU training pilot. The ordinary full `contacts-v1` validation loss should remain directly comparable to exp117 (interesting below 2.8).

Results so far

A stateless prototype now maps every example index to a deterministic `(epoch, shard, slot)`, constructs documents directly from contact rows, and best-fit packs an entire shard into exactly 2,650 output slots. Overfull shards uniformly sample packed bins; underfull shards use zero-loss padding. Synthetic tests pass; no real data transfer or training run has started.